Visualization of Weightlifting Variables to Strength and Total Kg Predictions

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1. Figure 1

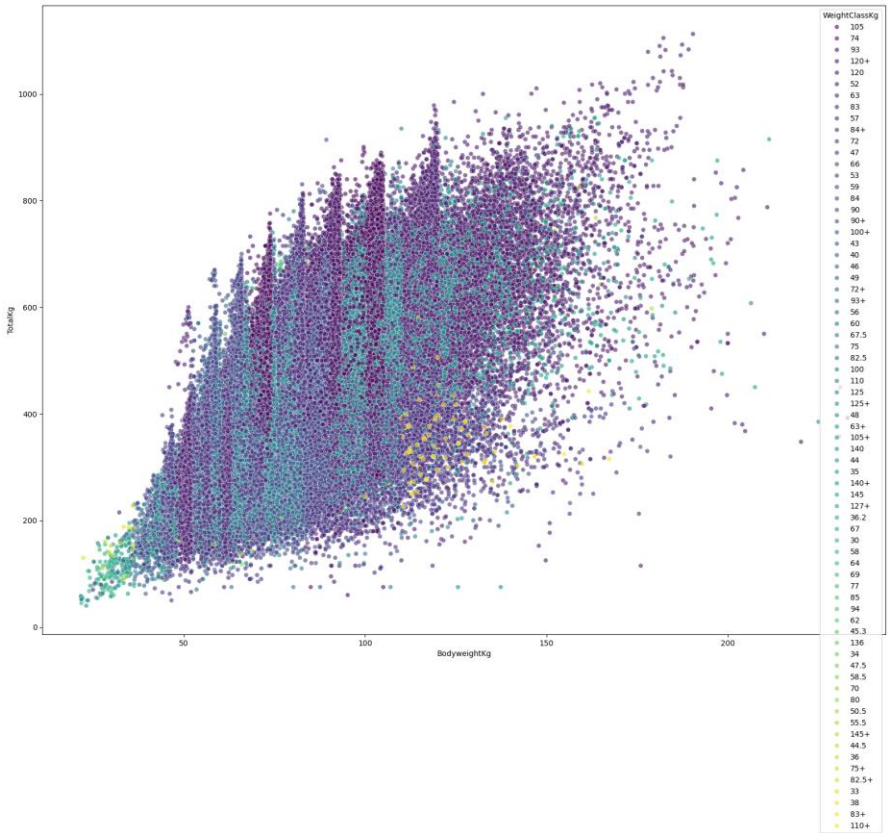

Figure 1 explanation:

Figure two displays a large scatter plot highlighting the broad linear correlation between one's total amount lifted and their body weight. This simple insight allows us to make a general conclusion that the more mass a person has, the more mass they will be able to lift.

Key Insights:

- Amount lifted is primarily determined by Sex
- Age plays less of a factor in ability to lift weight than body weight and sex
- The relationship between amount of weight lifted for each of the three lifts is strong

Figure 2:

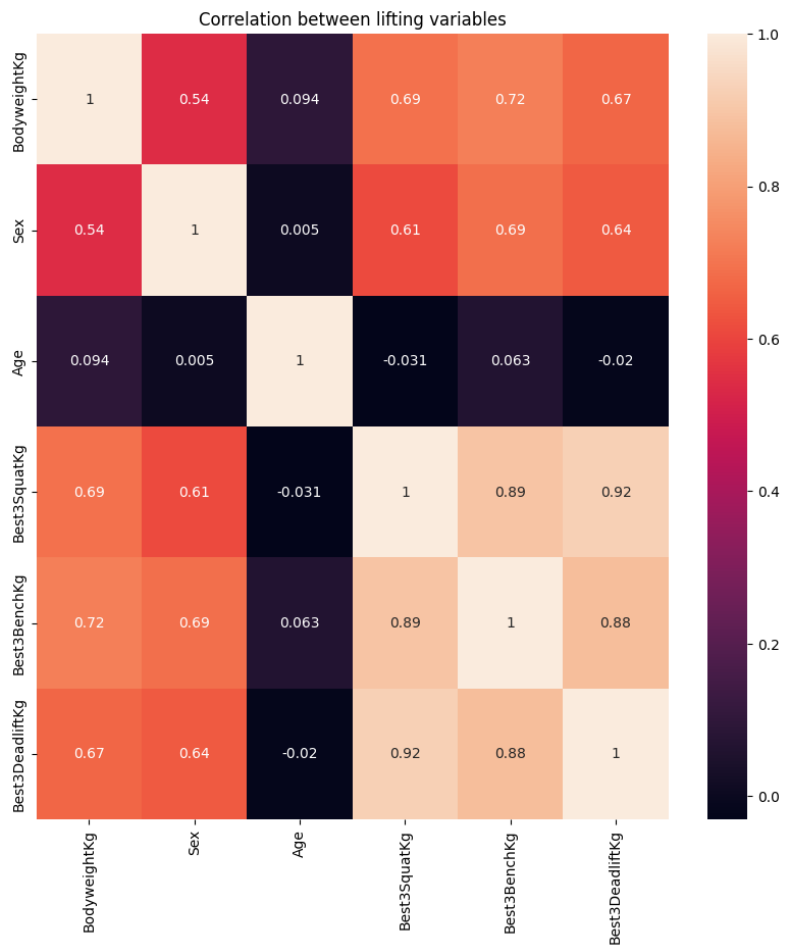

Figure 2 Explanation:

In this figure, we see the relationship between the variables that contribute to one's total weight lifted in a powerlifting meet:

- The axis represents the factors that were used from the data set to make this determination

- The color represents the strength of the relationship between variables
    - high correlation is represented by the palest color and low correlation is represented by the darkest.
- The leading determinant of one's meet total is their body weight.
- The lift with the highest correlation to one's body weight is bench press, while deadlift has the lowest correlation out of the three
- Sex has the second-highest correlation to determining how much someone can lift on all three lifts.
- High correlation between one life and another shows positive correlation between squat, bench, and deadlift

Data and Methods:

The data for this was obtained through the 'openpowerlifting' CSV file, which contains 145,3278 rows of information. This data is shared with the public and uses data from all International Powerlifting Federation to compile the set. To refine my data extraction, I cleaned the data to get rid of unused variables such as lifters' names, where they competed, and other null or repeated data. For generating the figures, I used Pandas, Seaborn for the scatterplot, and Matplotlib. These libraries were used to read the data and display it in a way that was efficient for understanding, such as a correlation-heatmap and scatter plot.

Significance:

The significance of this data is the messaging it conveys for people interested in starting an athletic hobby, such as powerlifting: that there are no barriers to entry based on uncontrollable factors. It can be daunting to start something new due to the uncertainty of ability and resources, but this data tells a different story. The story told by this information is that the responsibility of one's personal factors that might seem like deterrents, such as weight, age, etc. Likely playing less of a role than one might think, or has the potential to benefit.

Github Link: https://github.com/KarinaFeindt/InfoSciFinal